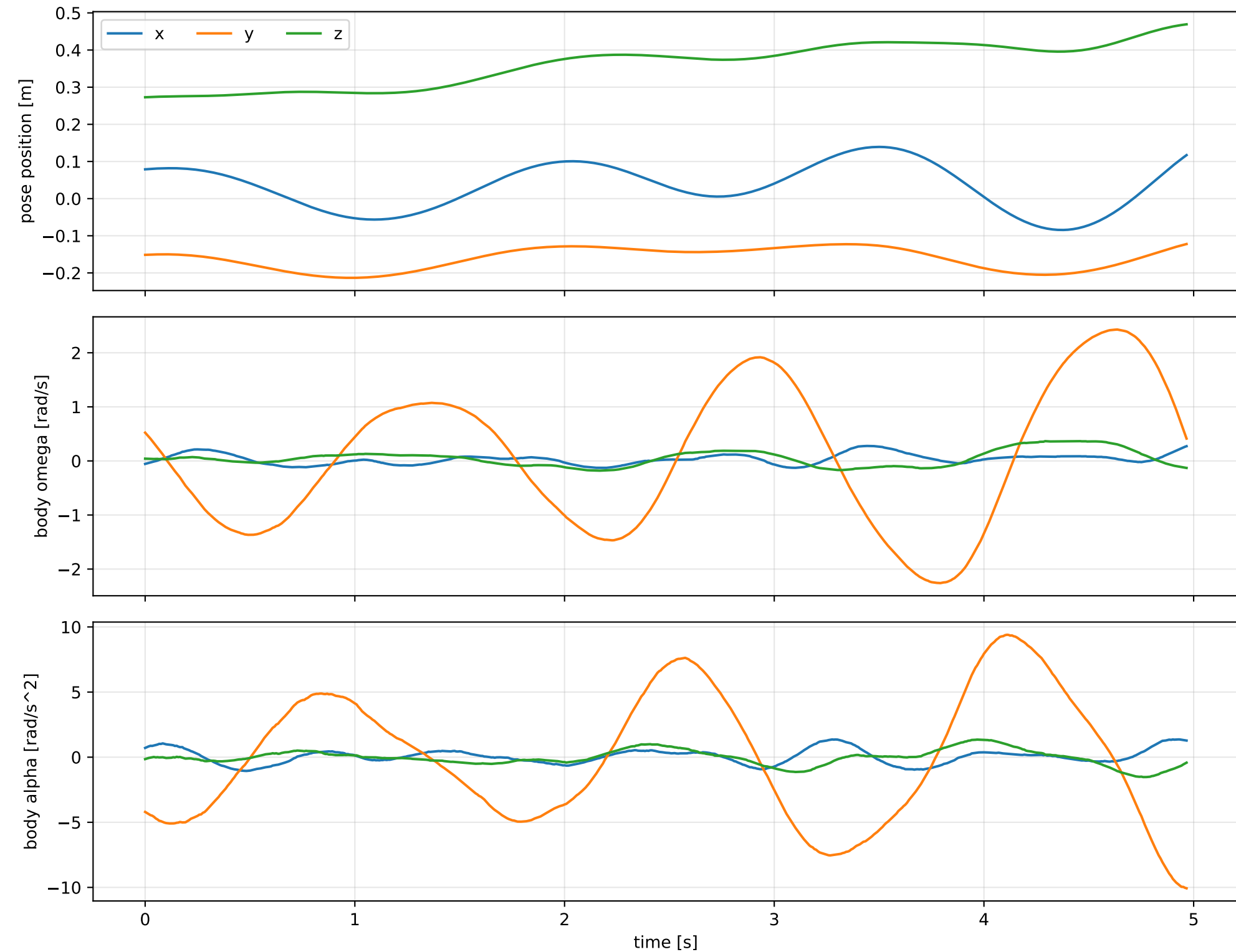
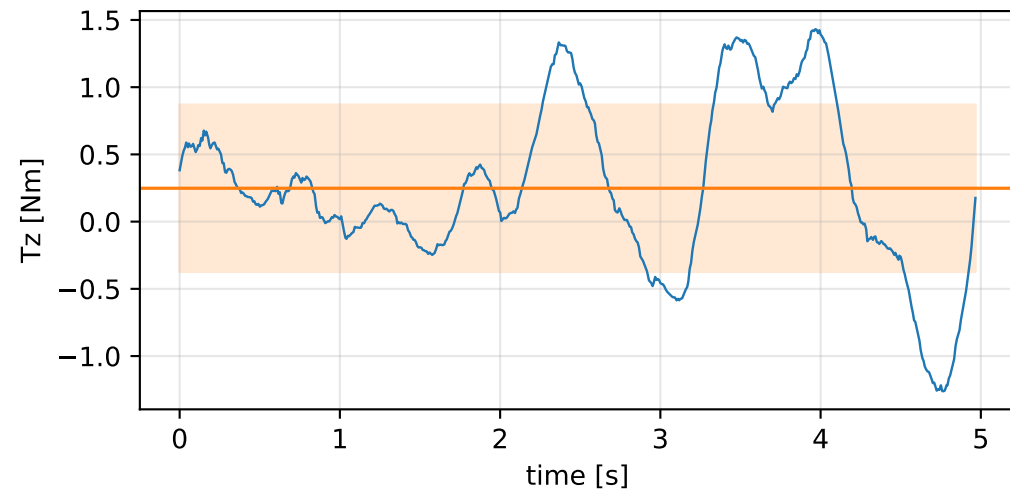
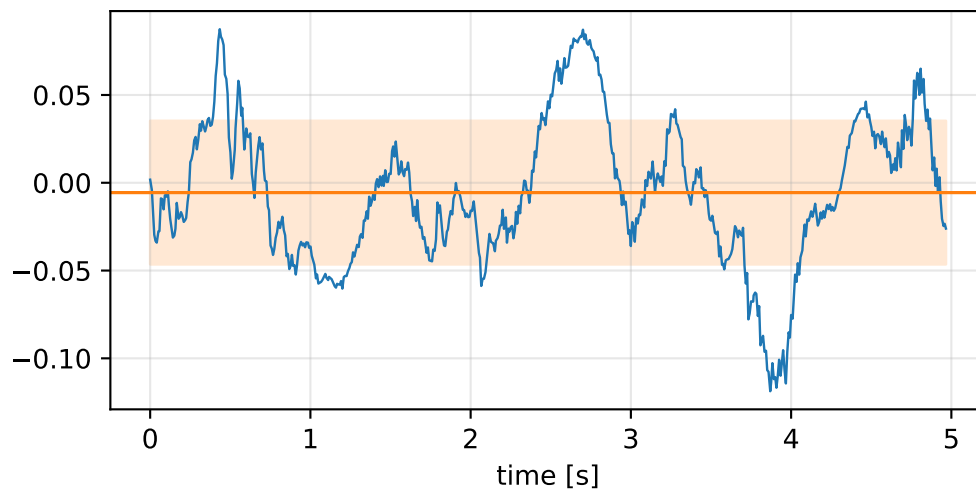
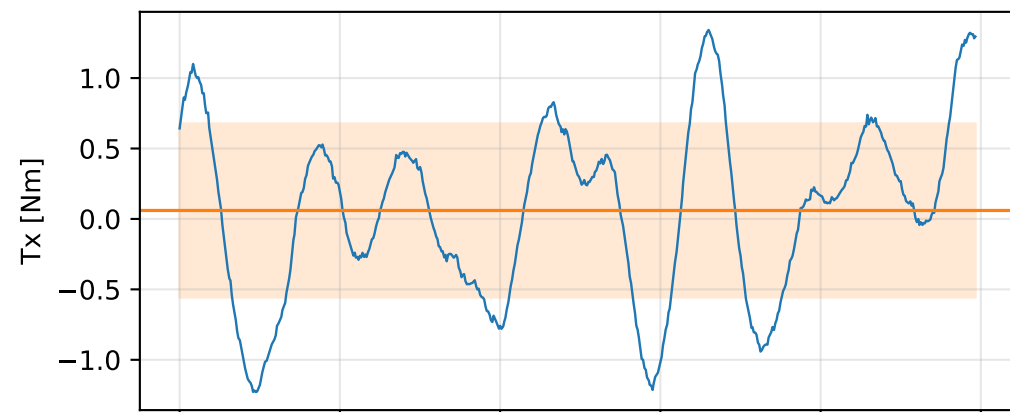
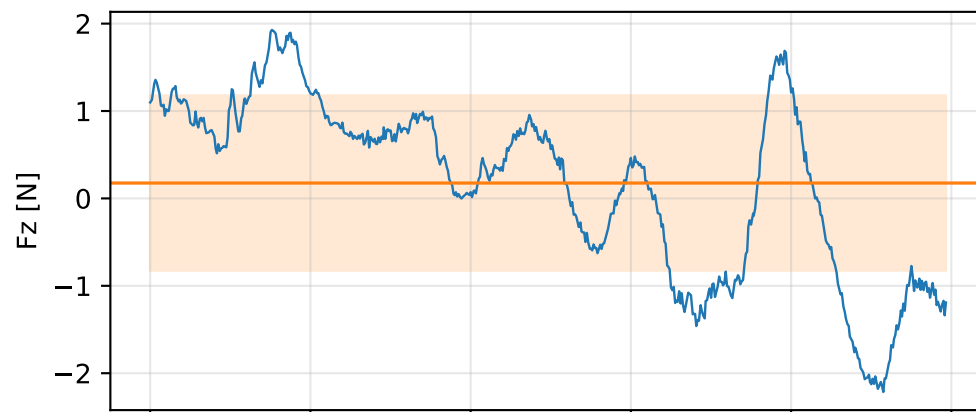
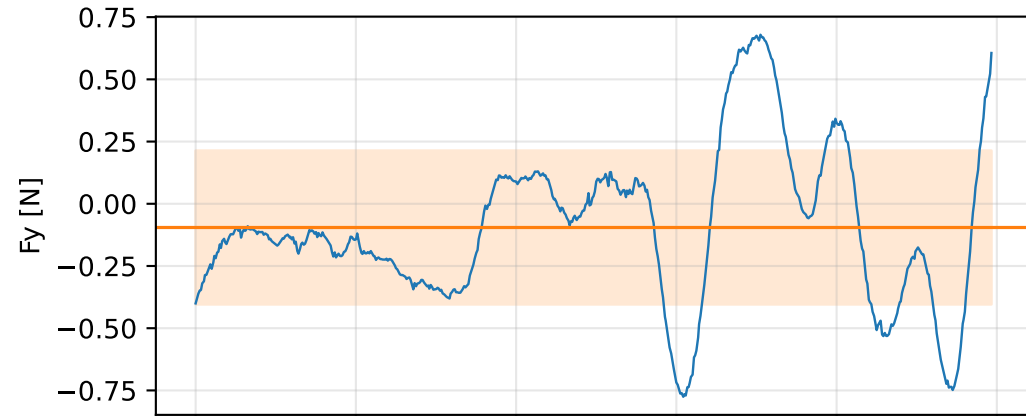
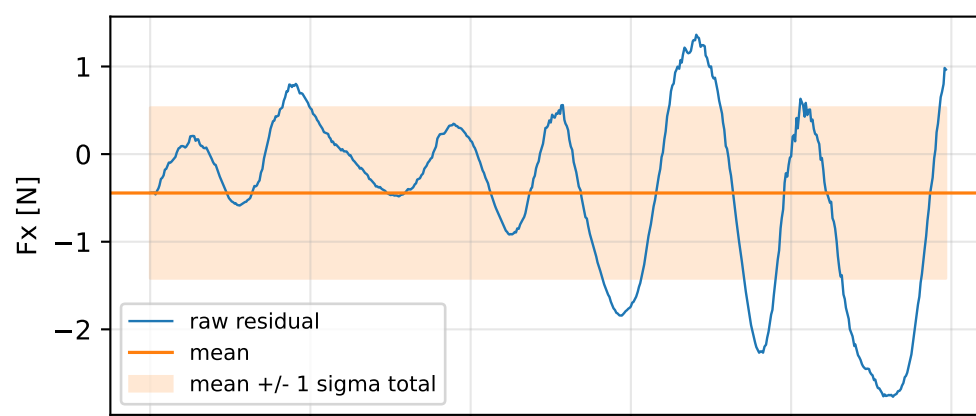


inertia_over_mass_yy_nominal: SG trajectory and derived motion



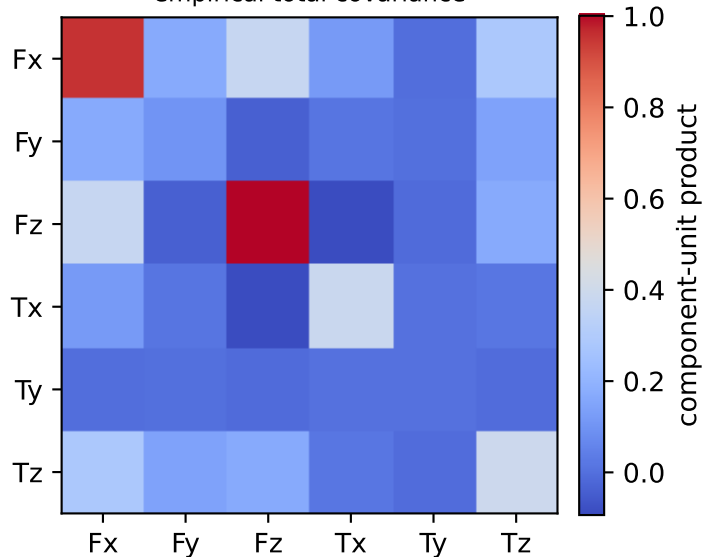
inertia_over_mass_yy_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



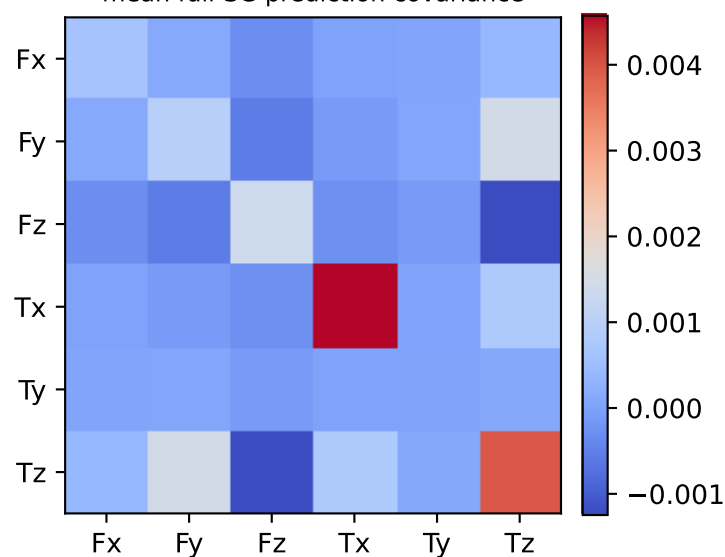
vector RMS about zero: force 1.5097 N, torque 0.91132 Nm; component std about mean: force [0.9743 0.3088 1.0004], torque [0.6161 0.0409 0.6207]

inertia_over_mass_yy_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [1.235e-03 1.753e-02 2.593e-01 3.427e-01 7.339e-01 1.446e+00]

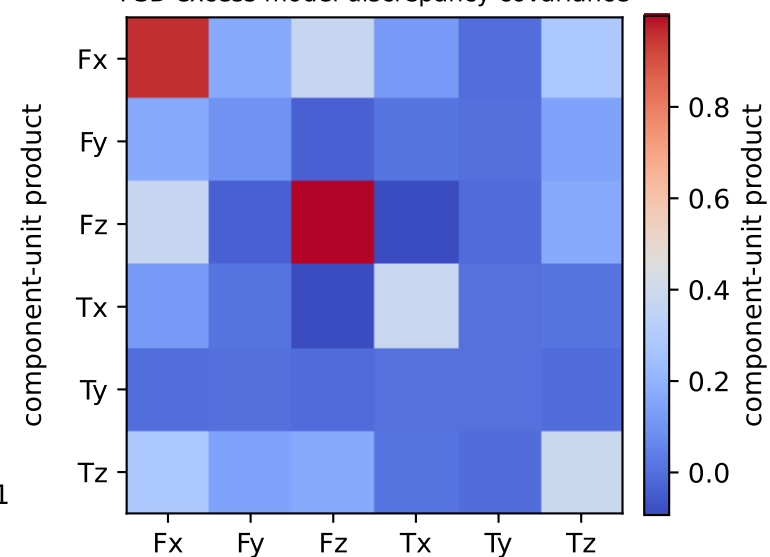
empirical total covariance



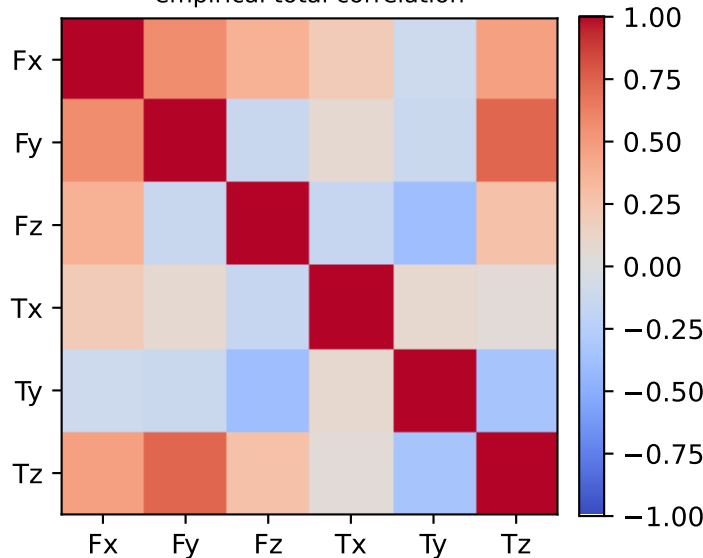
mean full-SG prediction covariance



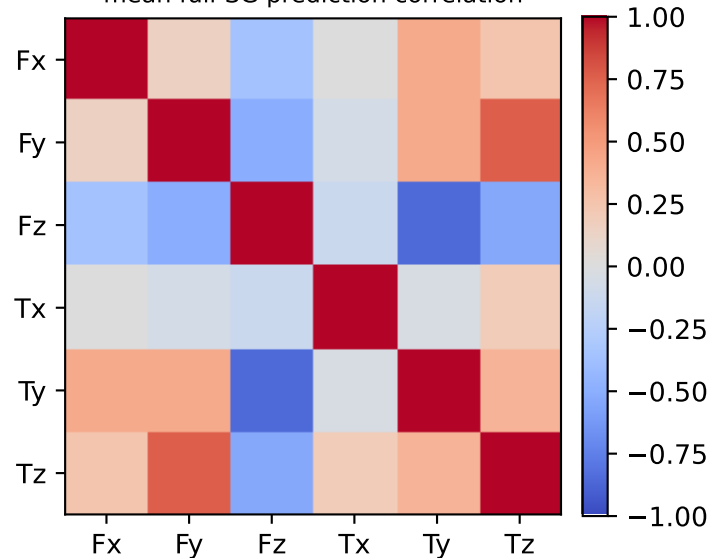
PSD excess model discrepancy covariance



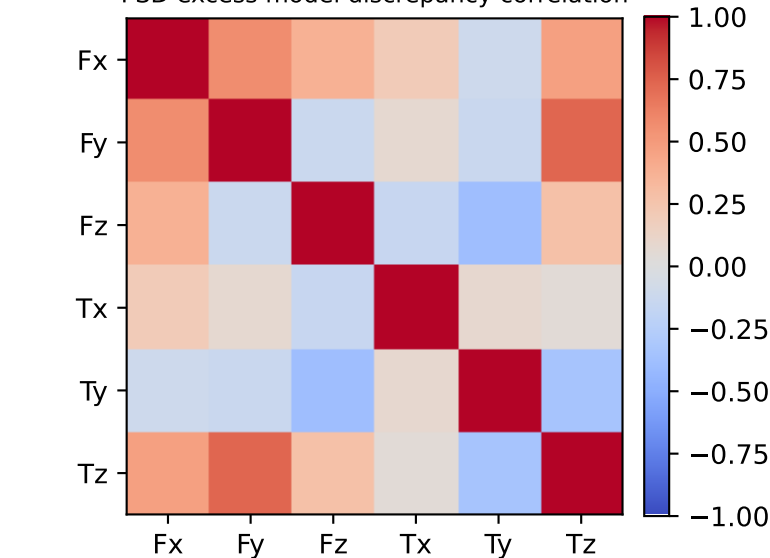
empirical total correlation



mean full-SG prediction correlation

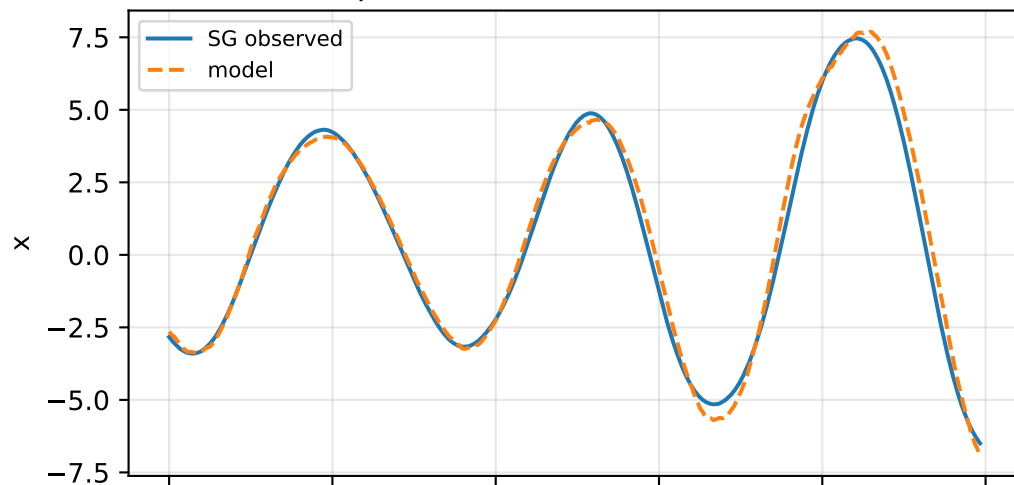


PSD excess model discrepancy correlation

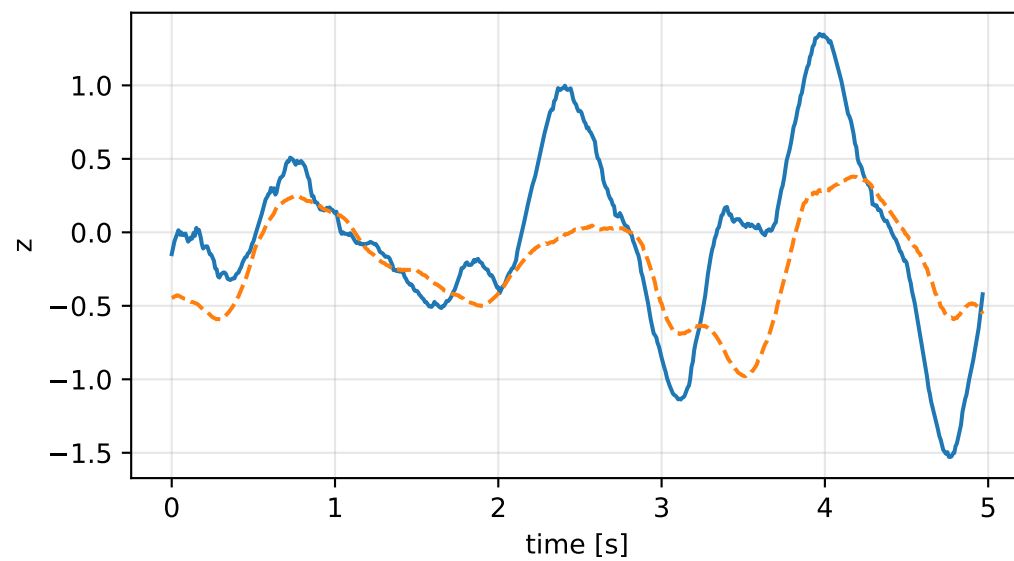
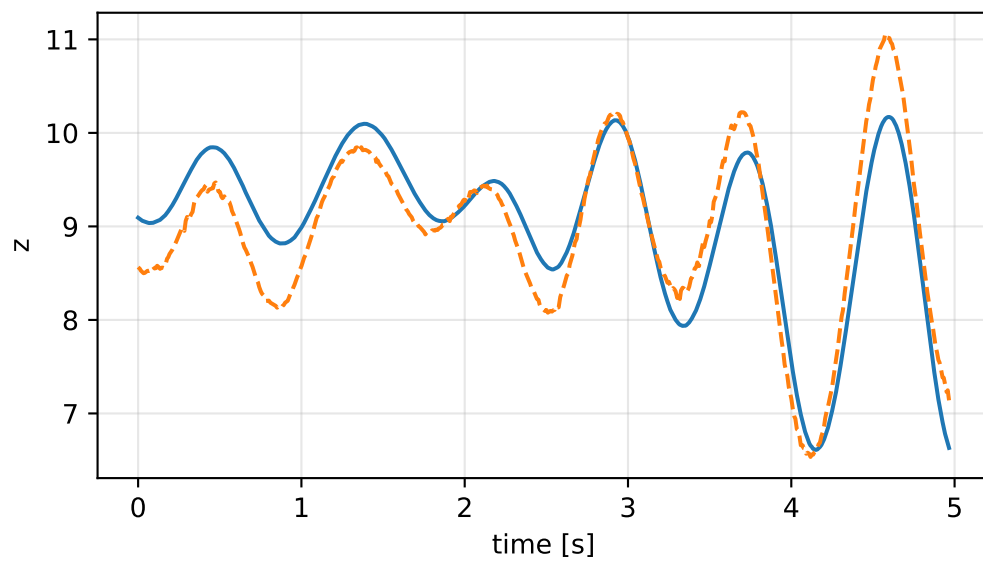
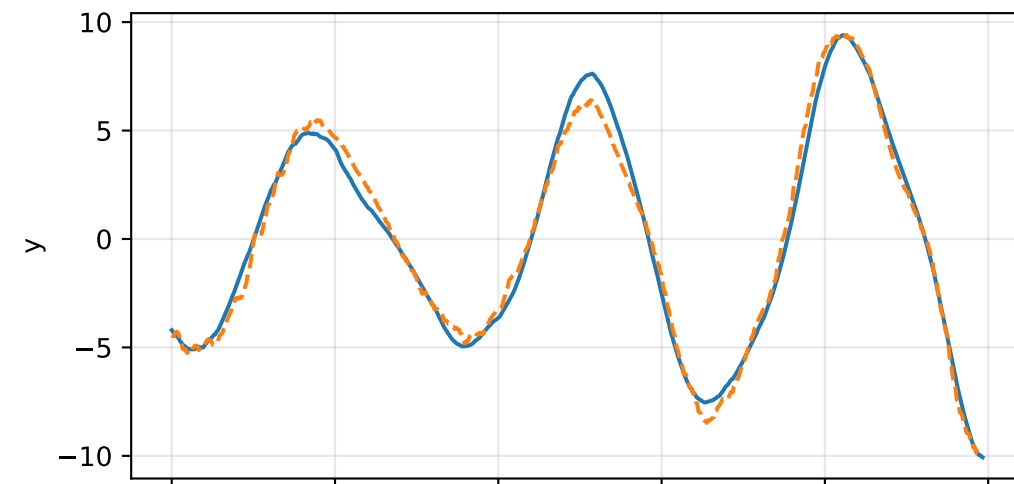
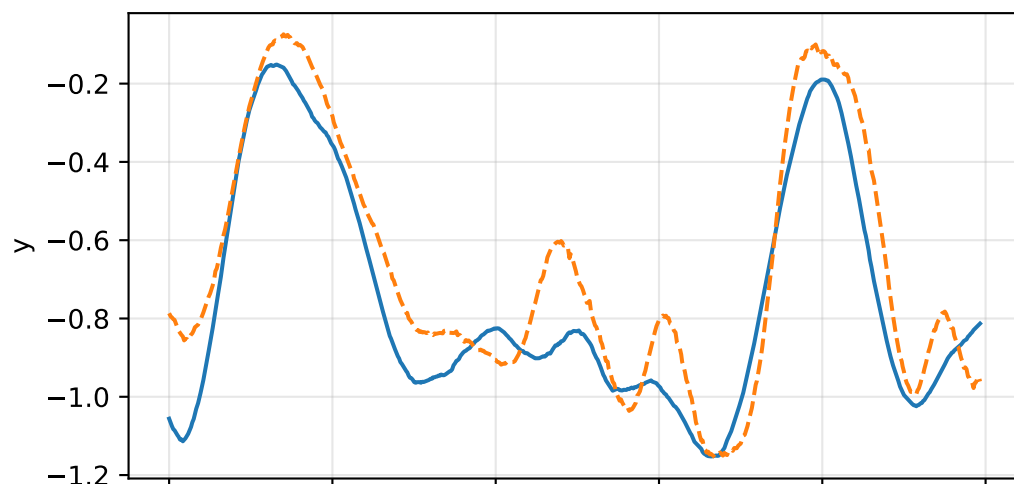
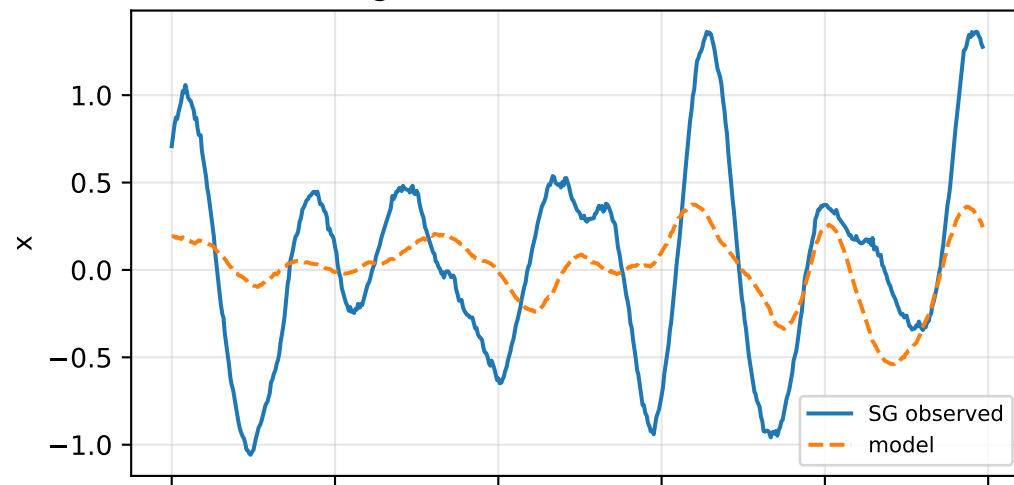


inertia_over_mass_yy_nominal: acceleration residual objective

specific acceleration [m/s^2]

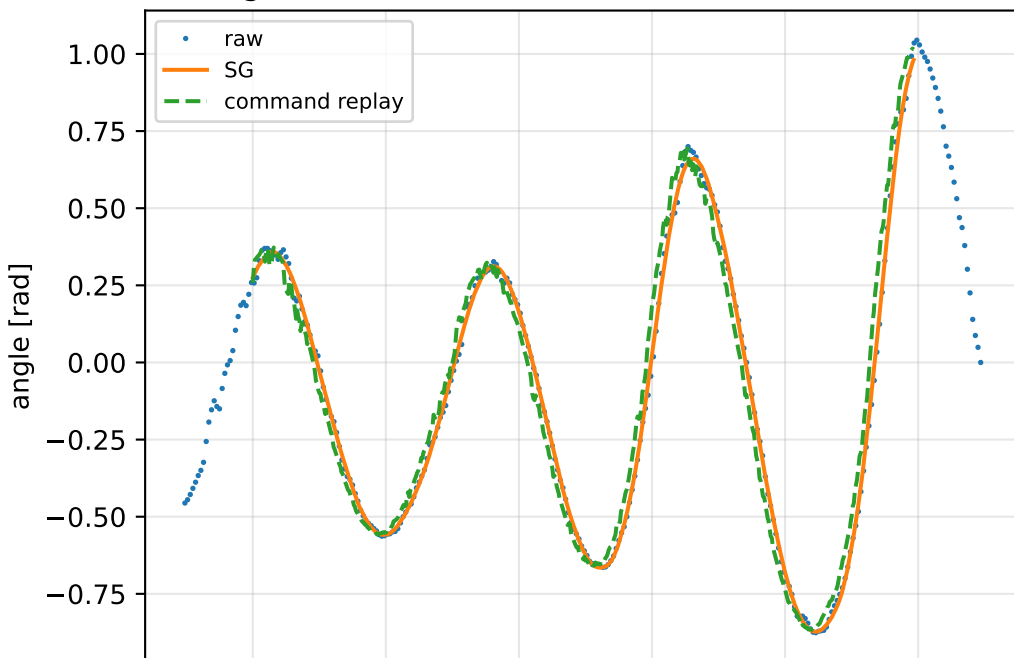


angular acceleration [rad/s^2]

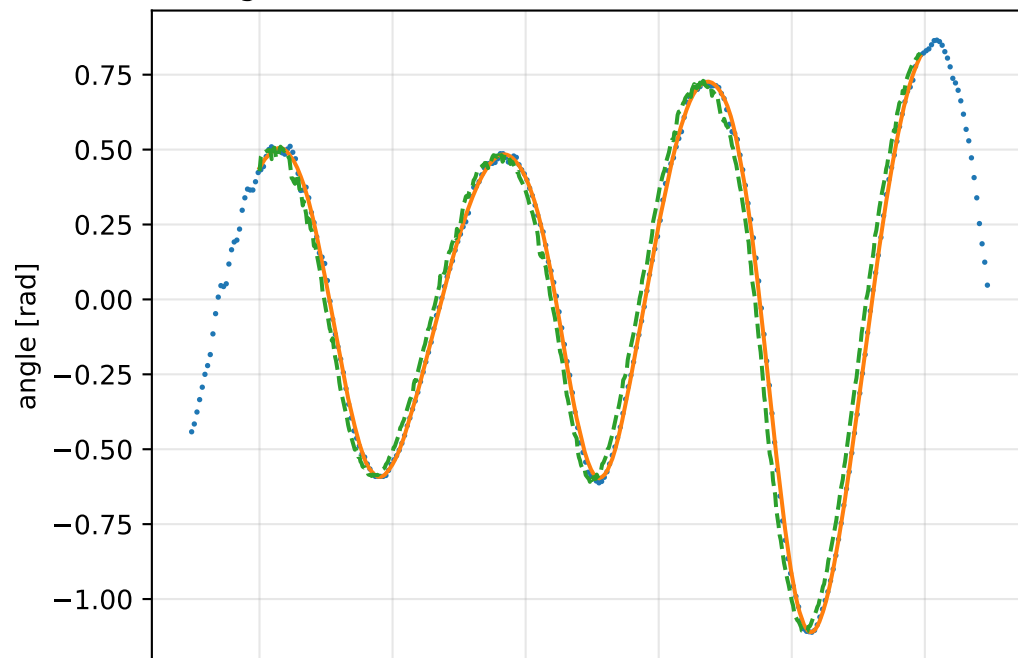


inertia_over_mass_yy_nominal: gimbal measurement audit (objective=measured_sg)

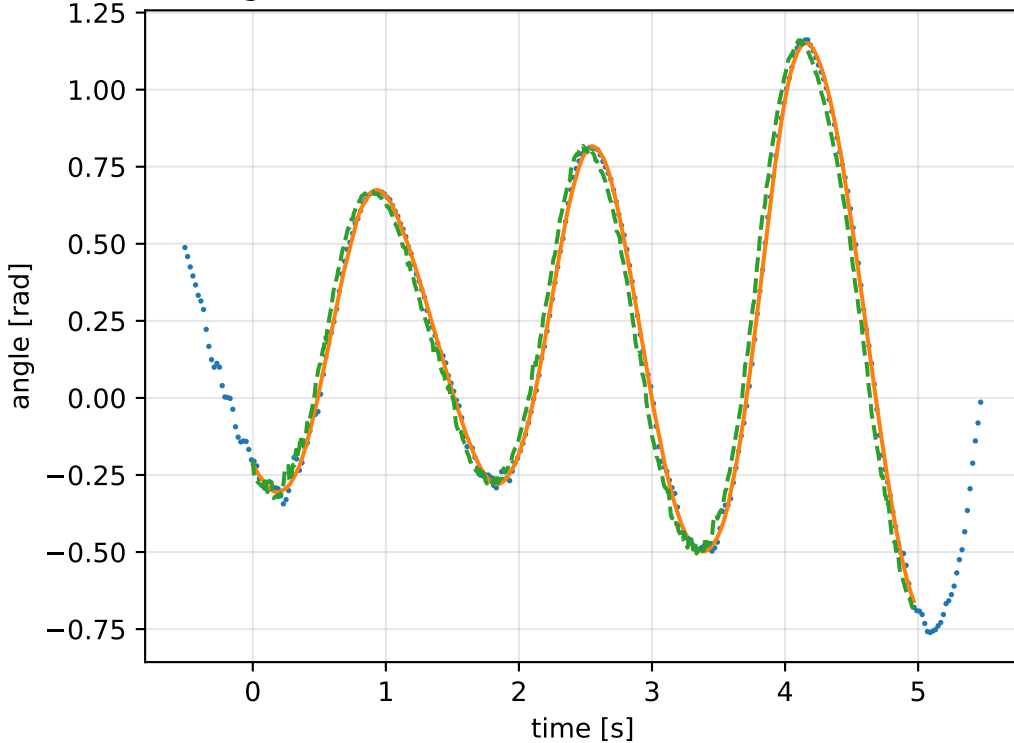
gimbal 1: RMSE=0.0759, max=0.2095 rad



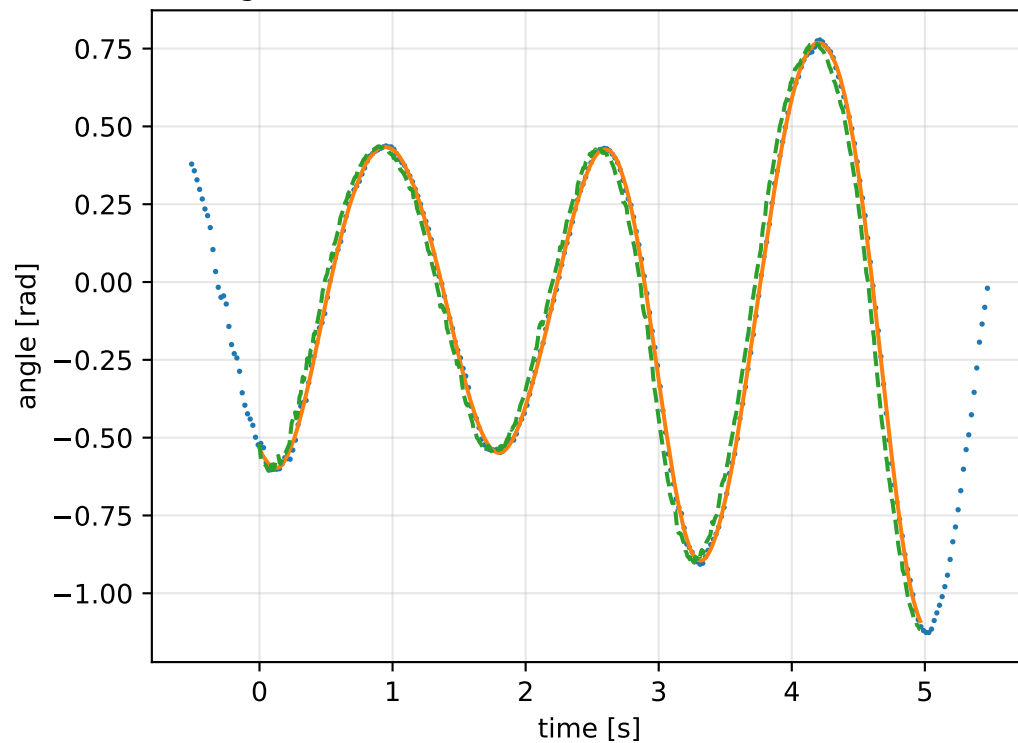
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

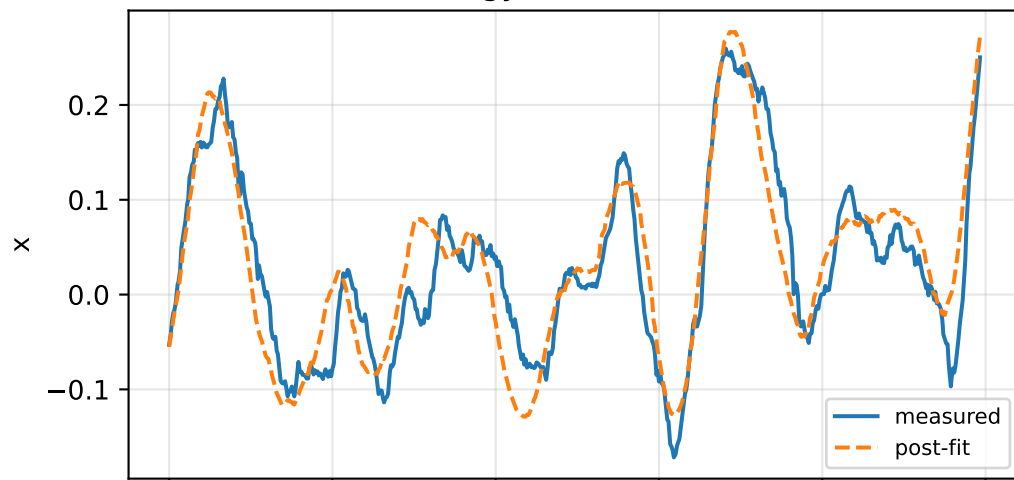


gimbal 4: RMSE=0.07131, max=0.1737 rad

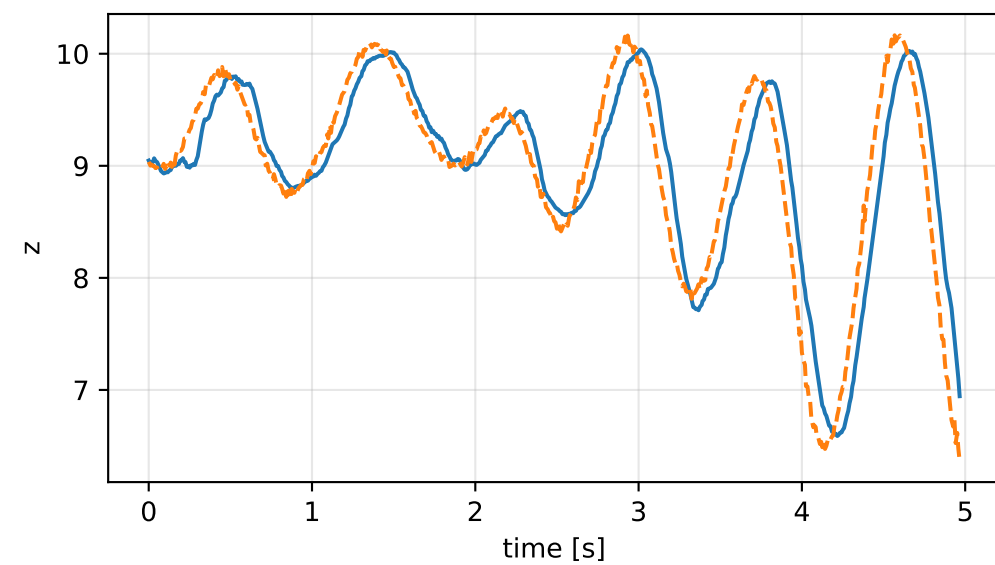
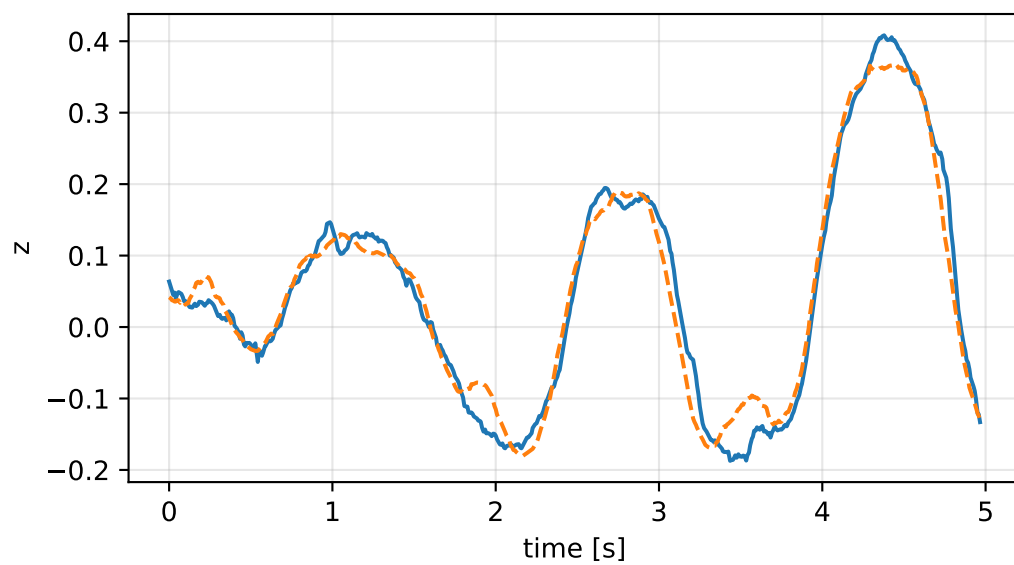
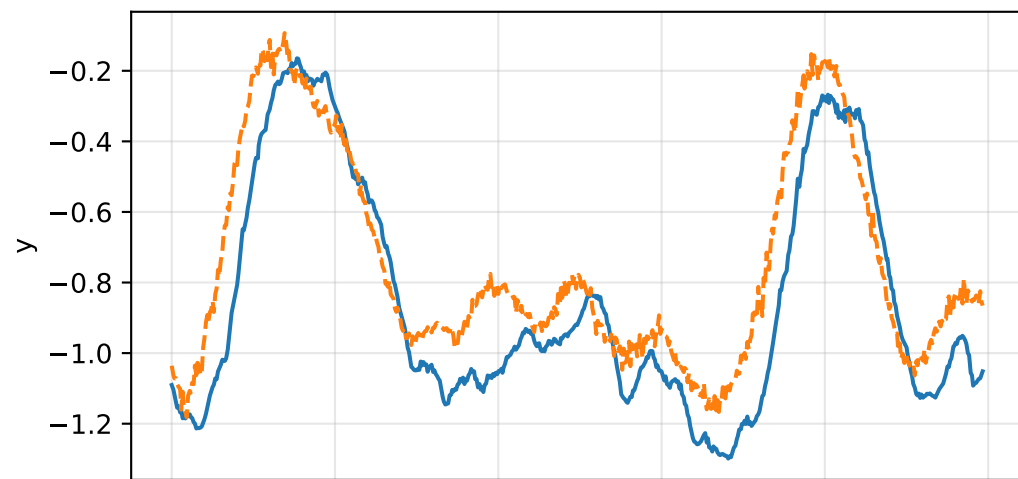
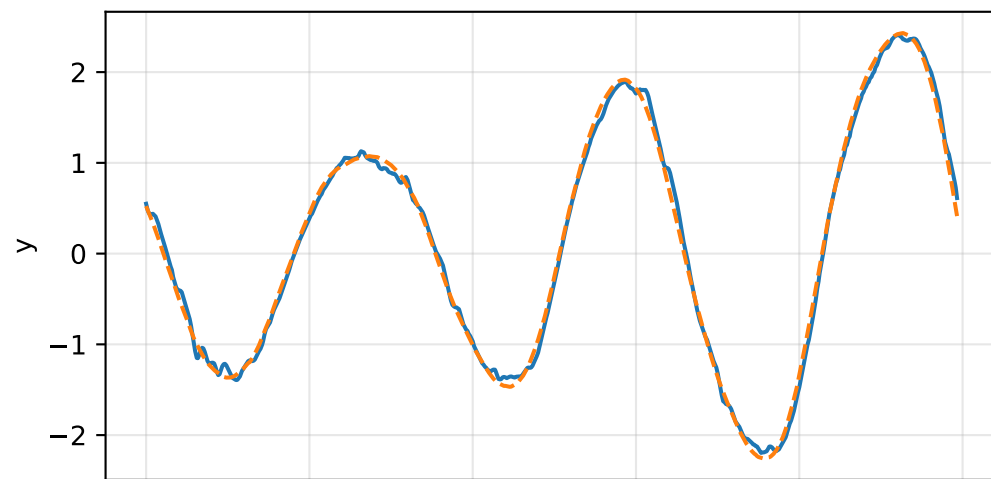
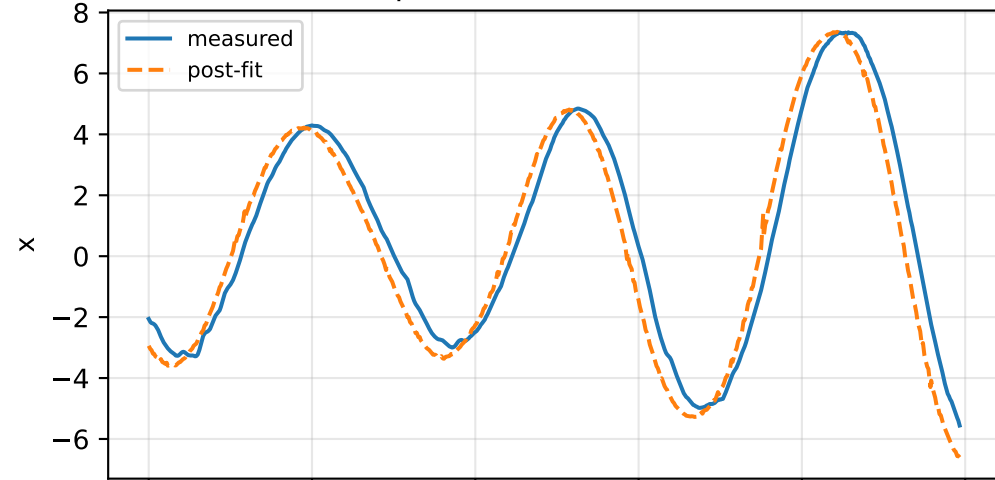


inertia_over_mass_yy_nominal: IMU comparison (diagnostic only)

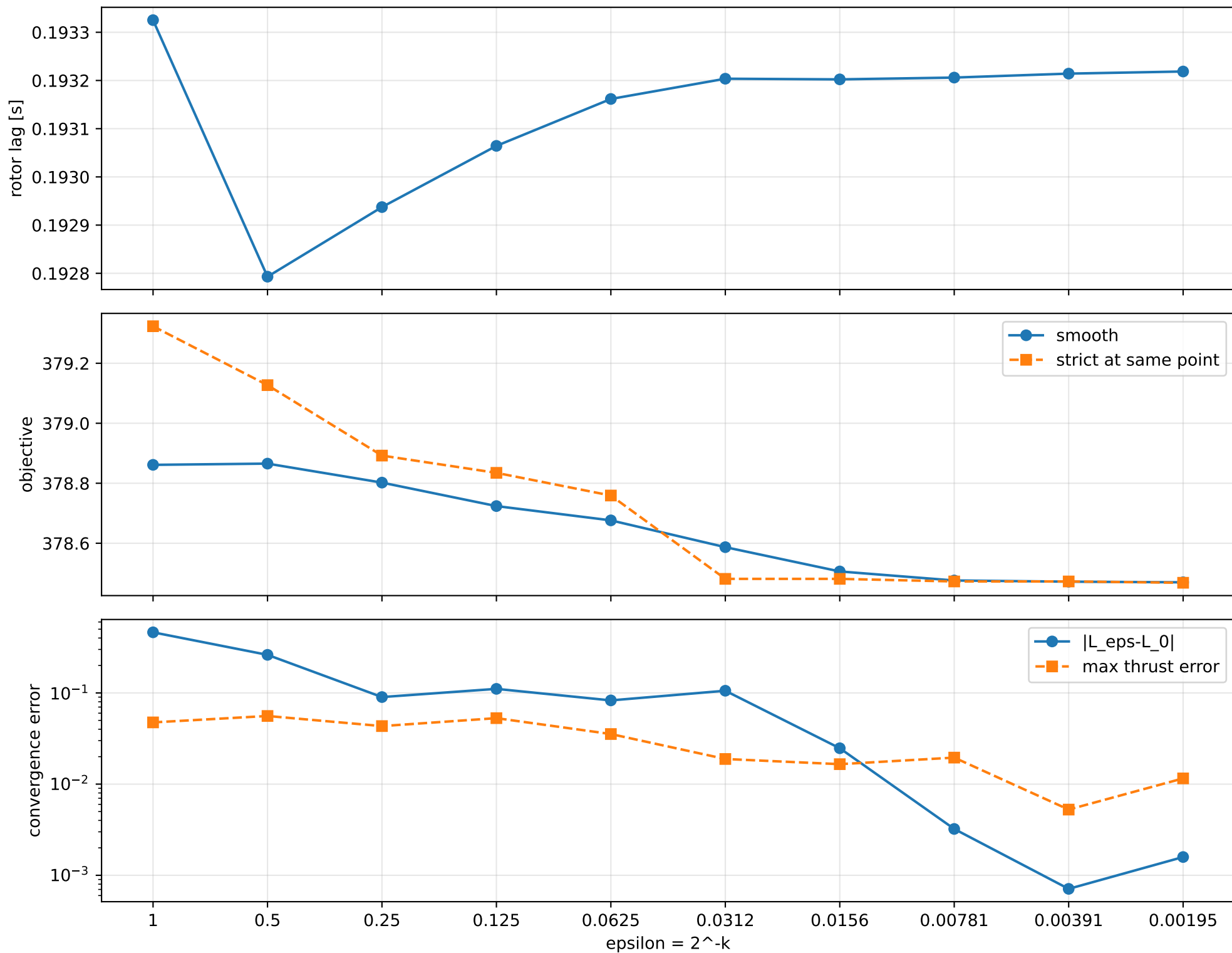
gyro [rad/s]



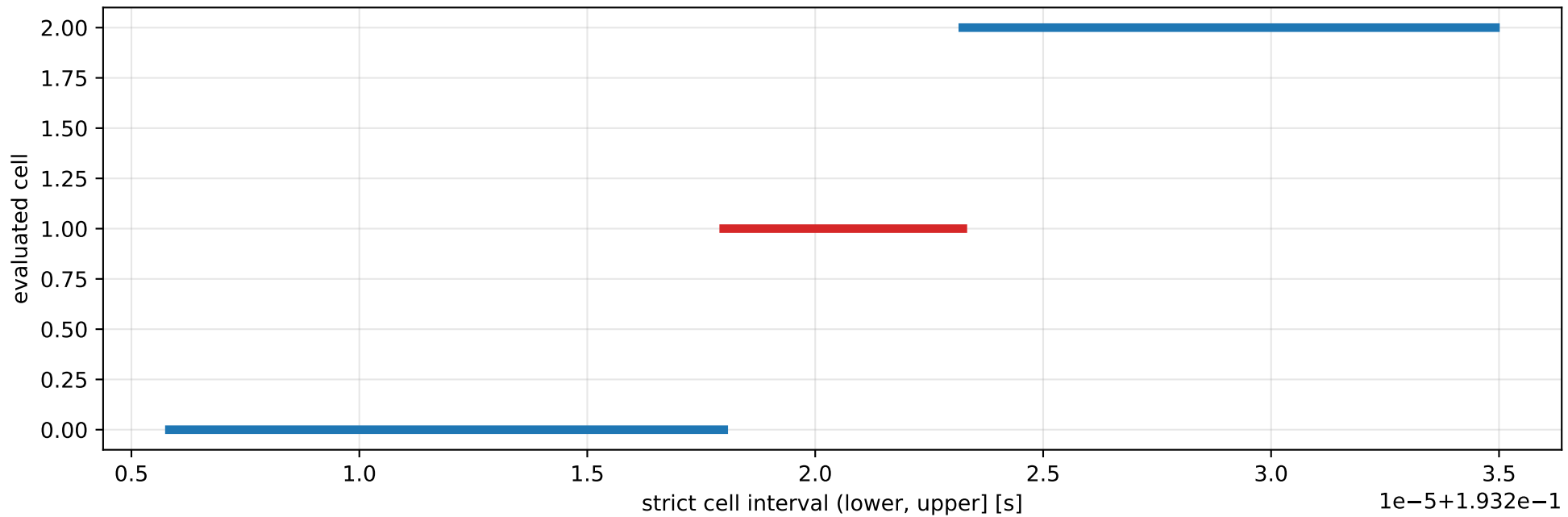
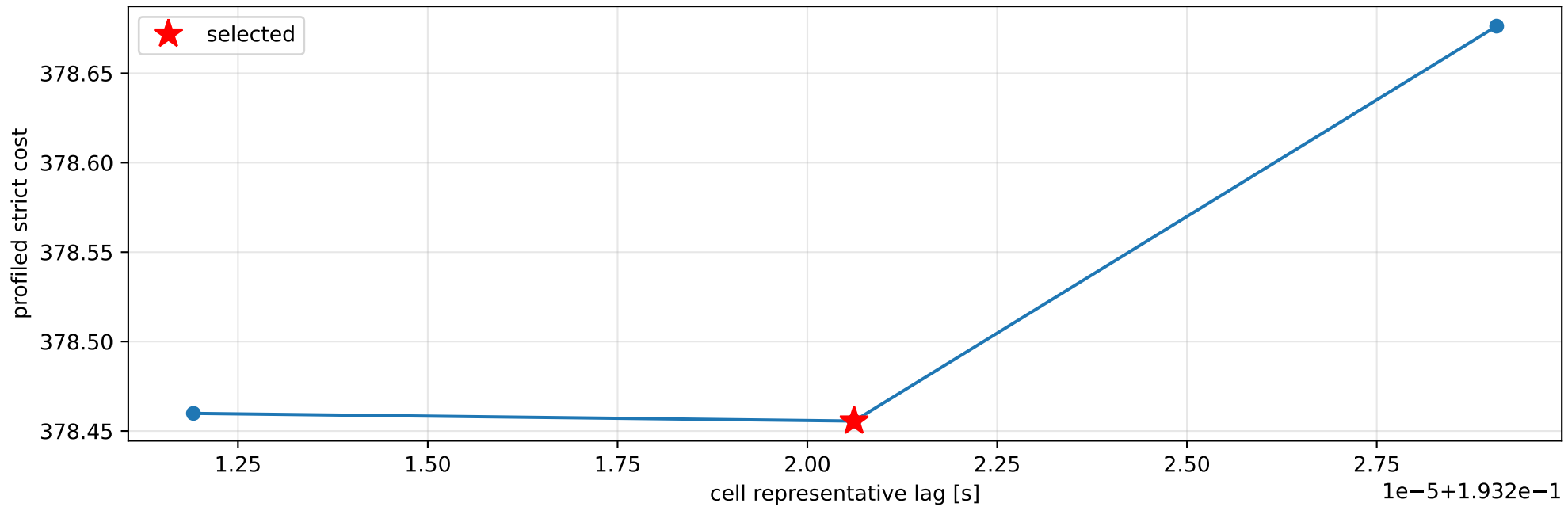
specific force [m/s^2]



inertia_over_mass_yy_nominal: smooth-to-strict continuation



inertia_over_mass_yy_nominal: exact strict-ZOH lag cells



selected (0.193217993, 0.193223238] s; final smooth 0.193218633 s; data boundary=False

inertia_over_mass_yy_nominal: parameter estimate

Case: inertia_over_mass_yy_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 2.82387071

inertia [kg m²]:

[[1.50305686 -0.01345812 -0.00470574]

[-0.01345812 0.07799716 -0.03401636]

[-0.00470574 -0.03401636 1.57879949]]

CoG body [m]: [0.17405531 -0.02908391 -0.02446369]

force effectiveness: [0.30796771 2.05993689 1.66247367 0.13097274]

scale-free J/m [m²]:

[[0.5322683 -0.00476584 -0.00166641]

[-0.00476584 0.02762065 -0.01204601]

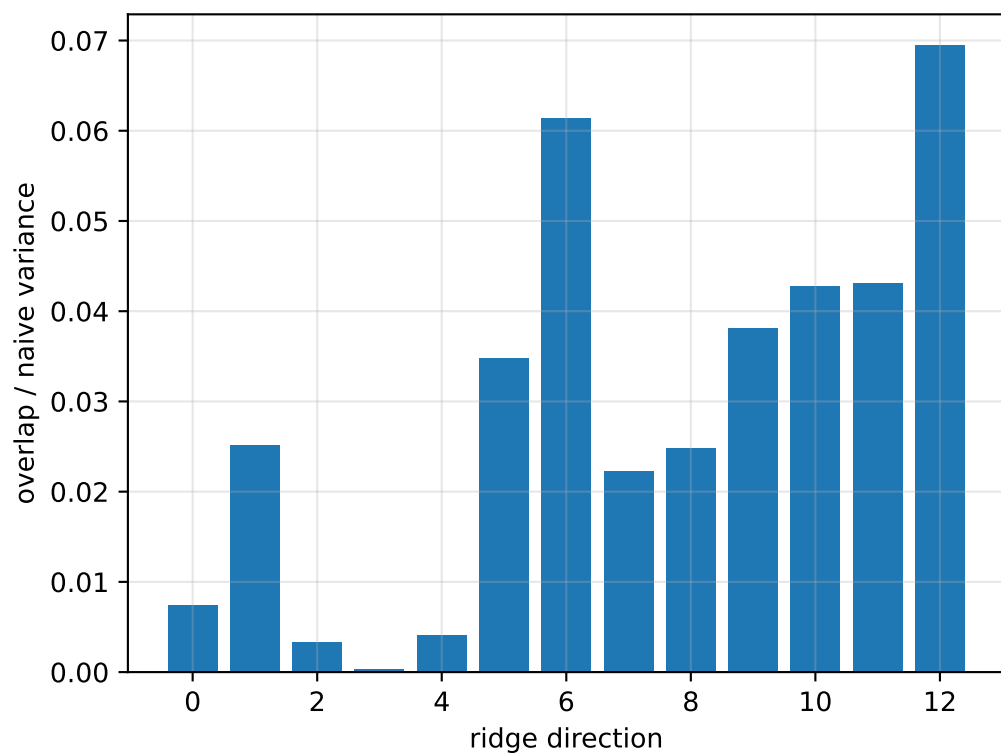
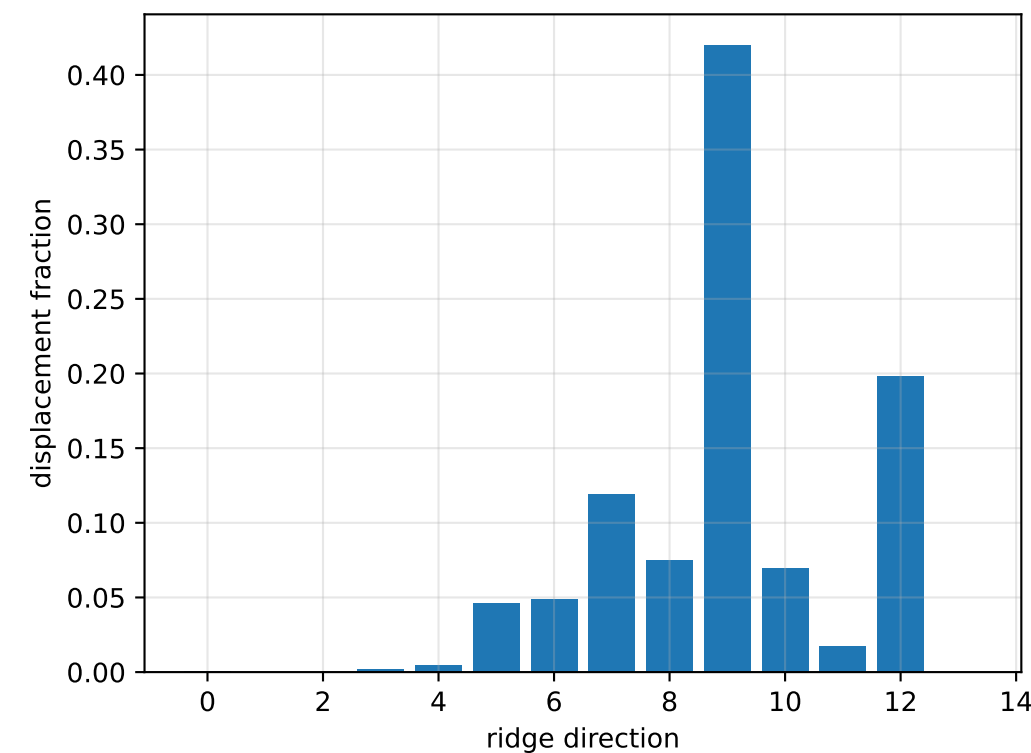
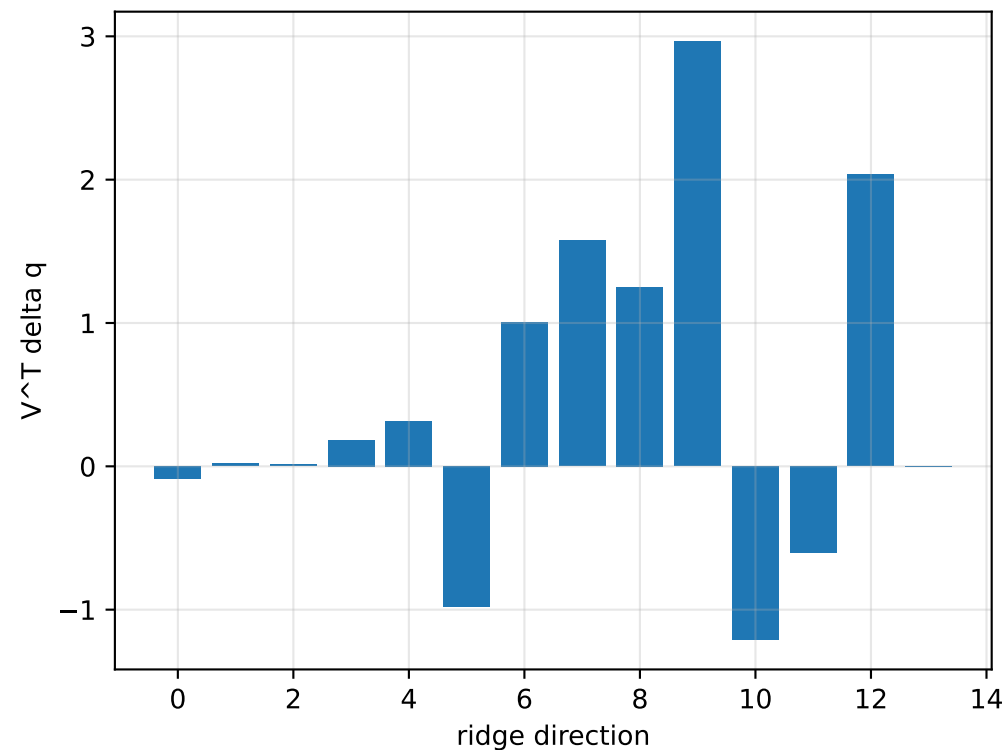
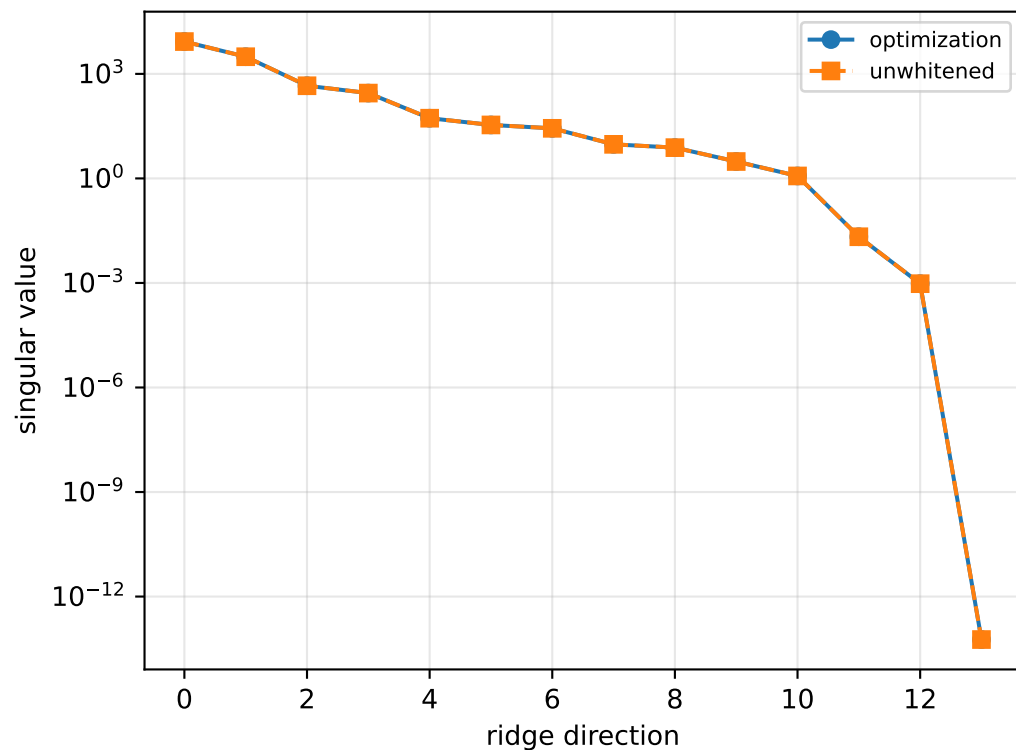
[-0.00166641 -0.01204601 0.55909057]]

scale-free f/m: [0.10905871 0.72947281 0.5887216 0.04638057]

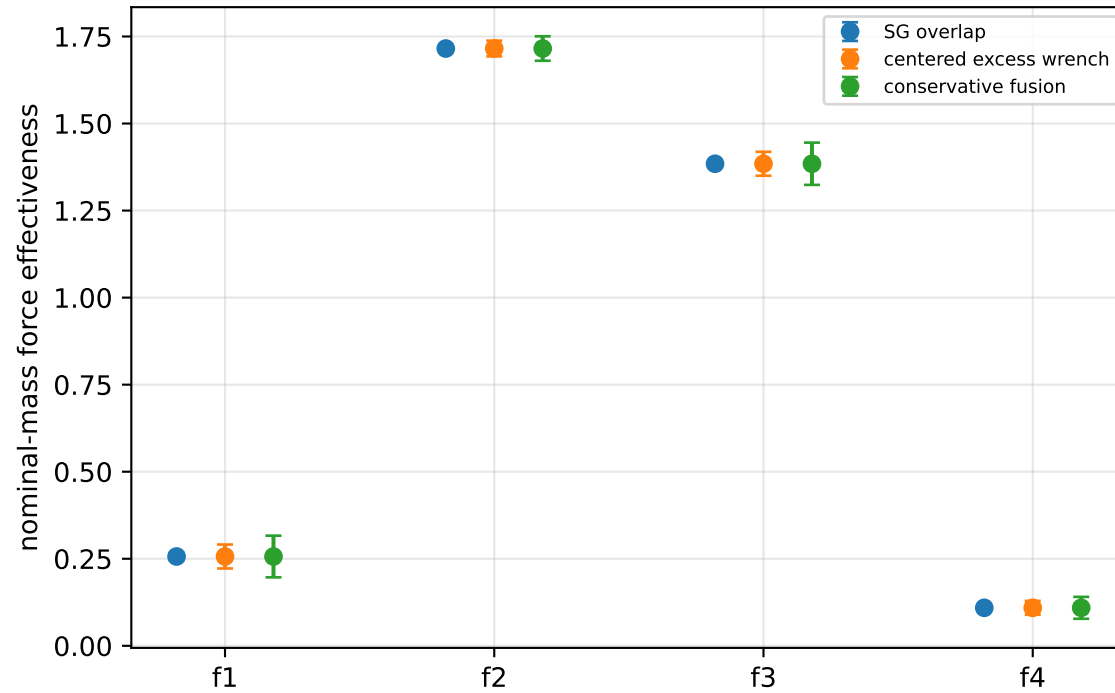
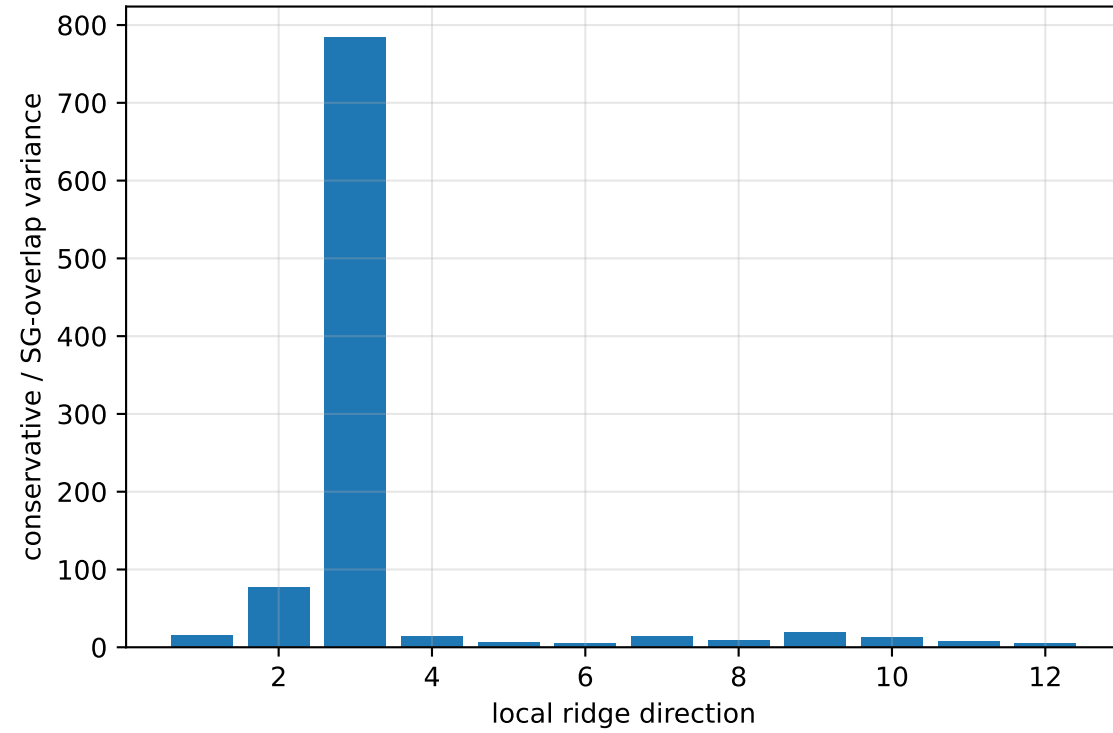
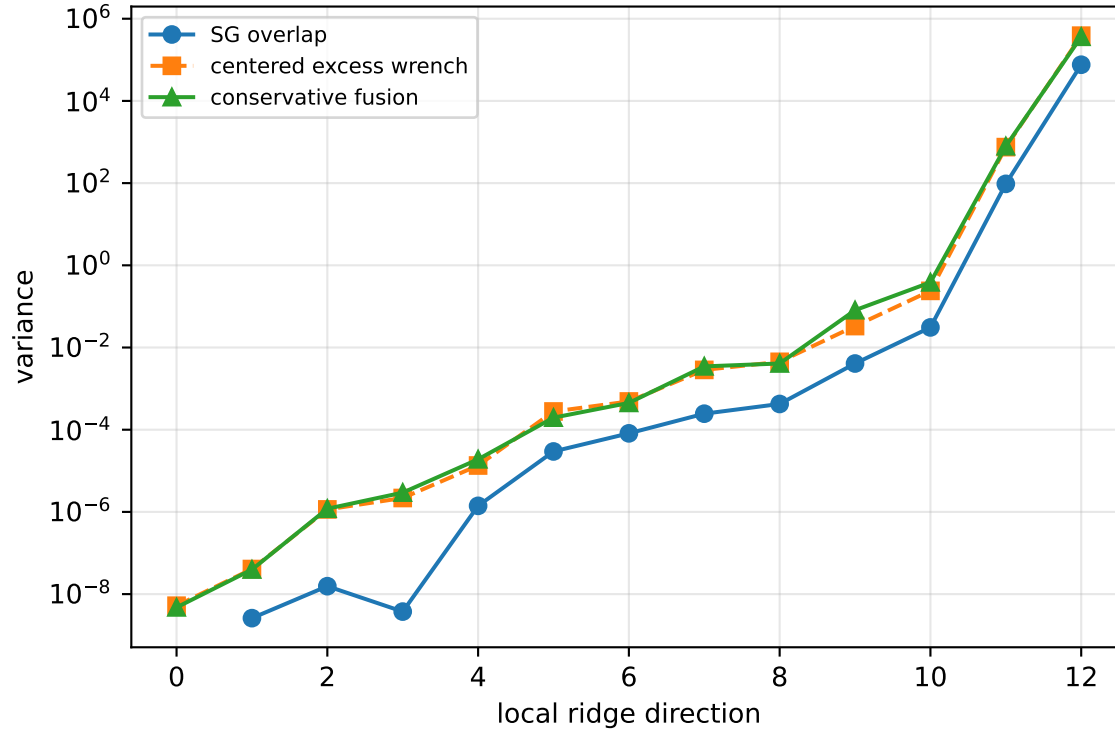
rotor lag cell: (0.193217993, 0.193223238] s

representative: 0.193220615 s

inertia_over_mass_yy_nominal: ridge spectrum (rank 13, nullity 1, ||v||=2.25e-13)



inertia_over_mass_yy_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions

$\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 35.56$
wrench-to-acceleration recovery max error = $3.33\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

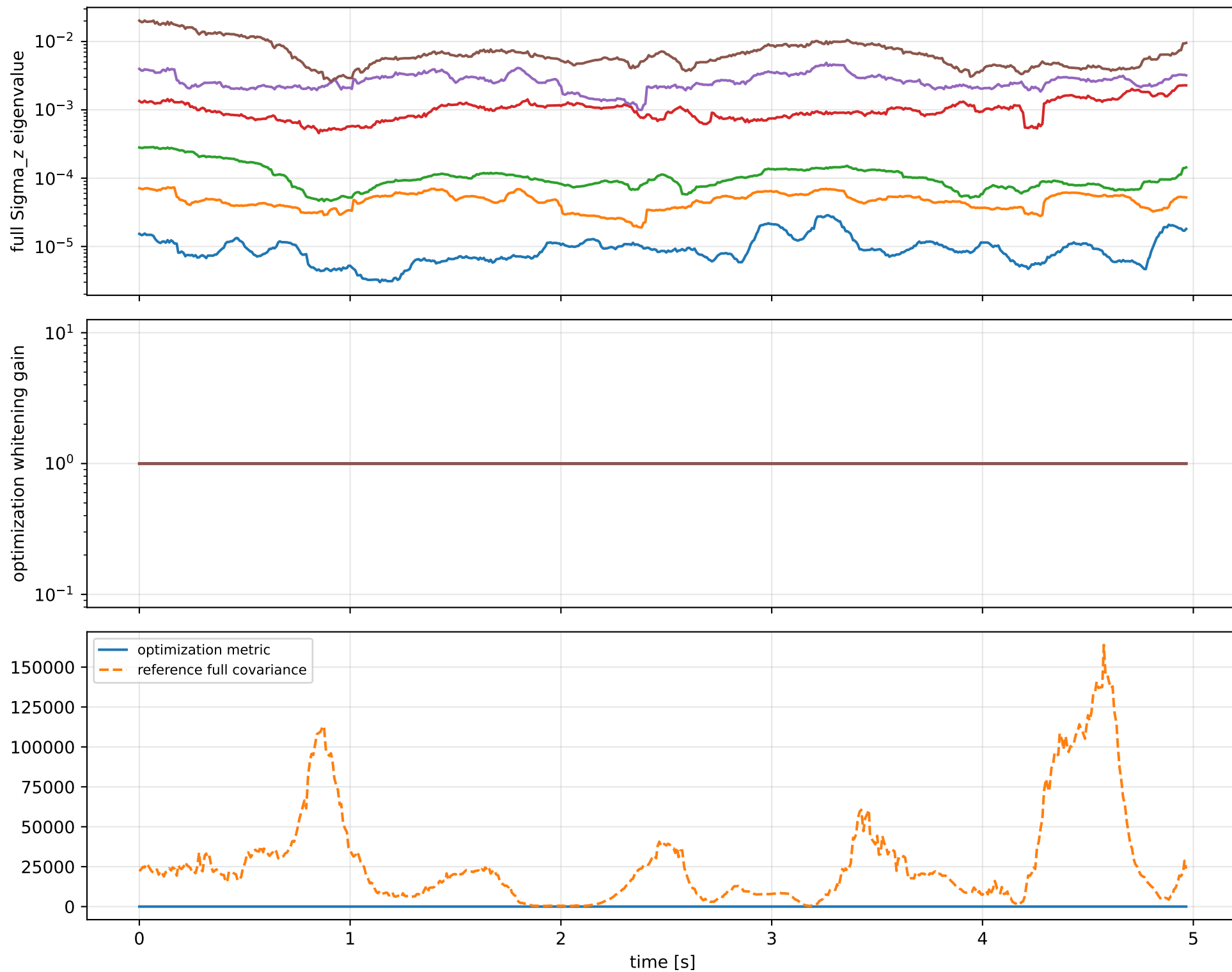
index 12: variance= $3.633\text{e}+05$, ratio=4.781
 $\log_{\text{mass}}=+0.136$; second_moment_diag_1=-0.0949
second_moment_diag_2=+0.0856; second_moment_diag_3=-0.671
second_moment_offdiag_12=-0.148; second_moment_offdiag_13=+0.582
second_moment_offdiag_23=+0.284; cog_x=+1.53e-06
cog_y=+1.29e-06; cog_z=+7.11e-07
log_force_eff_1=+0.136; log_force_eff_2=+0.136
log_force_eff_3=+0.136; log_force_eff_4=+0.136

index 11: variance=789.8, ratio=8.231
 $\log_{\text{mass}}=+0.0179$; second_moment_diag_1=+0.481
second_moment_diag_2=-0.0164; second_moment_diag_3=-0.553
second_moment_offdiag_12=+0.147; second_moment_offdiag_13=-0.634
second_moment_offdiag_23=+0.194; cog_x=+3.49e-05
cog_y=+3.09e-05; cog_z=-1.76e-05
log_force_eff_1=+0.0171; log_force_eff_2=+0.0179
log_force_eff_3=+0.018; log_force_eff_4=+0.0172

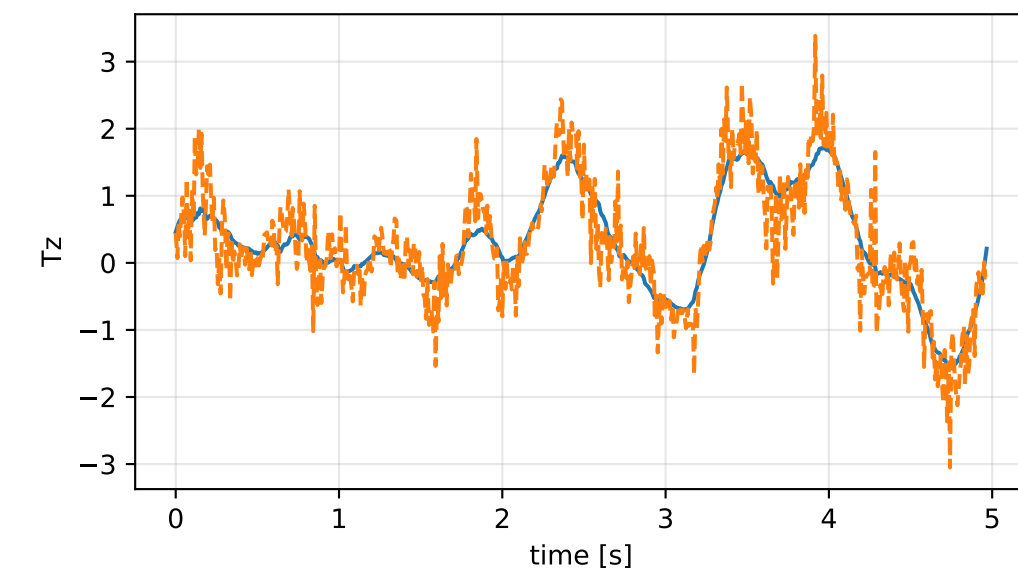
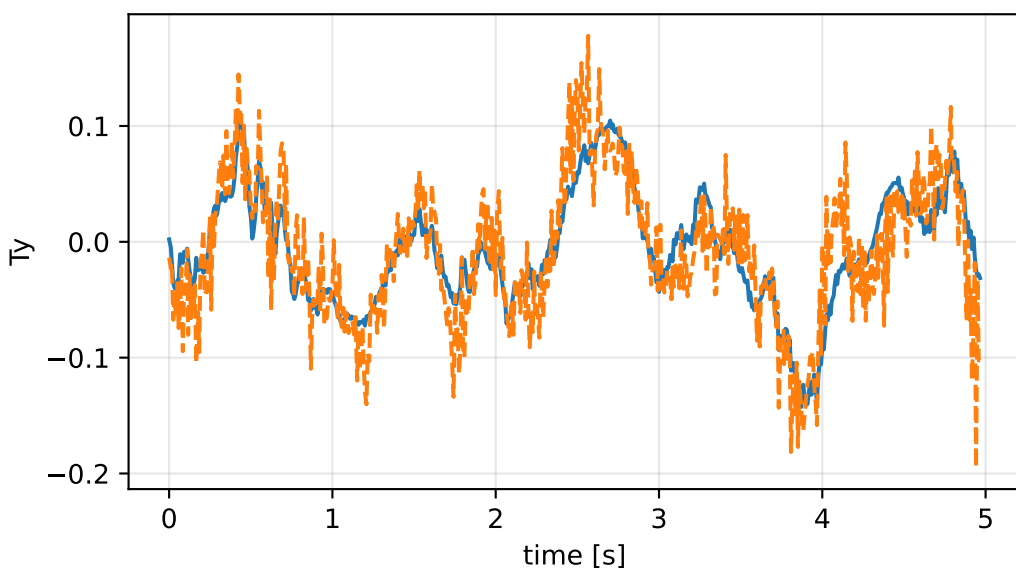
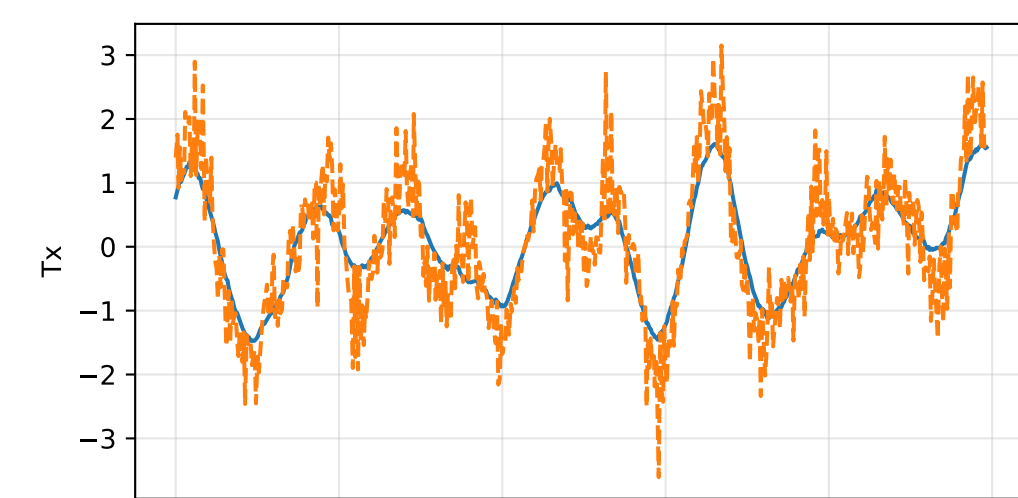
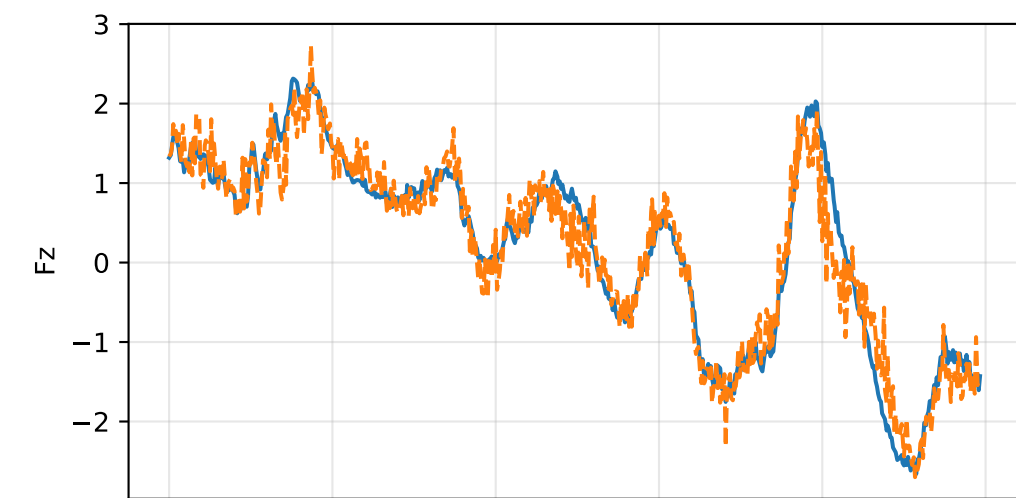
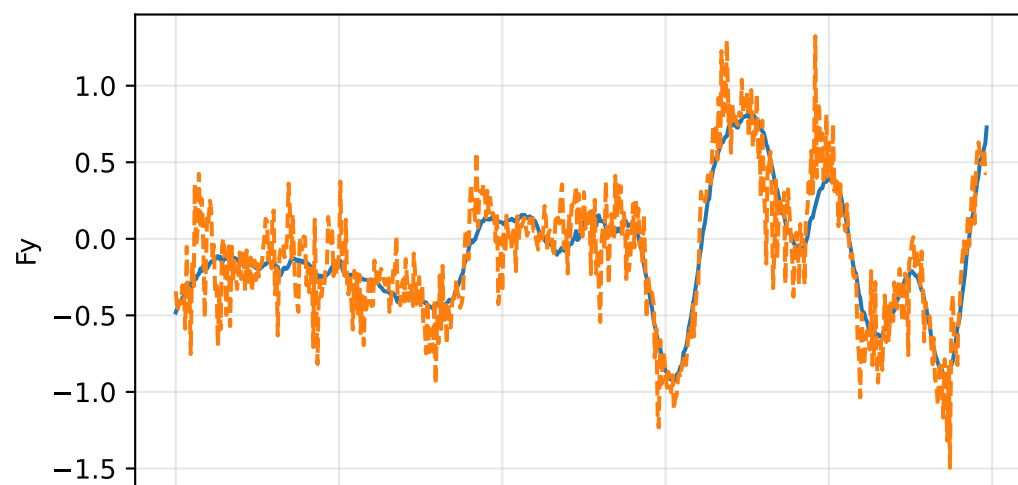
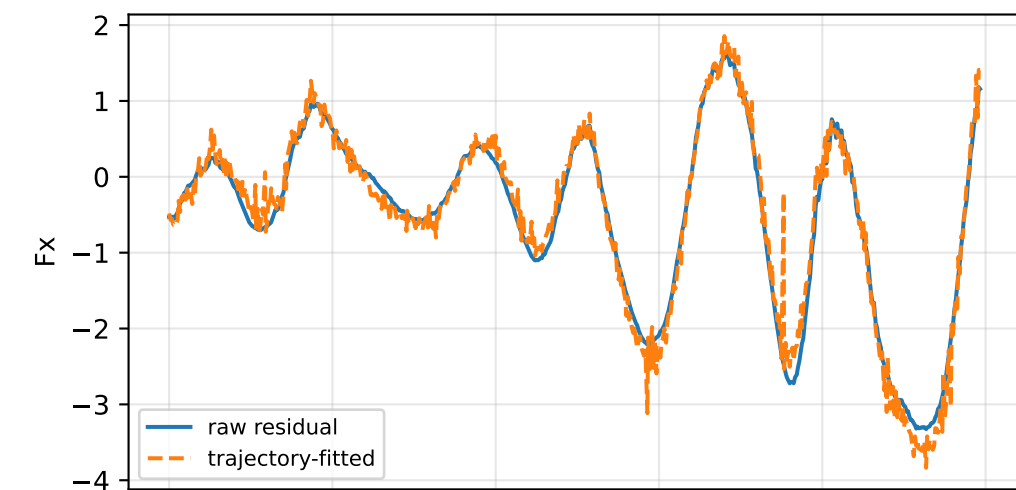
index 10: variance=0.3844, ratio=12.46
 $\log_{\text{mass}}=-0.0329$; second_moment_diag_1=+0.0862
second_moment_diag_2=+0.0961; second_moment_diag_3=-0.303
second_moment_offdiag_12=+0.224; second_moment_offdiag_13=+0.111
second_moment_offdiag_23=-0.886; cog_x=-0.00888
cog_y=-0.00414; cog_z=-0.000785
log_force_eff_1=+0.162; log_force_eff_2=-0.0448
log_force_eff_3=-0.0661; log_force_eff_4=+0.102

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_yy_nominal: optimization vs reference covariance



inertia_over_mass_yy_nominal: wrench / closure (force max $7.55\text{e-}15$, torque max $6.66\text{e-}16$)



inertia_over_mass_yy_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_yy_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma

SHA256: 62834c577333da807f8b968827a18fec3976eb136afd16ab23bddf481b6a6c3e

data objective: 378.455552583

prior objective: 4.71120676946e-06

total objective: 378.455557294

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_yy_nominal (inertia_over_mass_m2)

components: yy

target: [0.02762065]

std: [1.e-06]

final: [0.02762065]

error: [3.06959501e-09]

standardized residual: [0.0030696]

factor objective: 4.71120676946e-06